

\$37.25 in Boeing stock. What was it really worth?

Boeing buys Spirit AeroSystems and pays in its own shares: \$37.25 of Boeing stock per Spirit share, but only while Boeing trades between \$149.00 and \$206.94. The collar is priced as options on Boeing with its real implied volatility surface and followed every day for 17 months, to closing.

\$36.83

value of the offer
on signing day

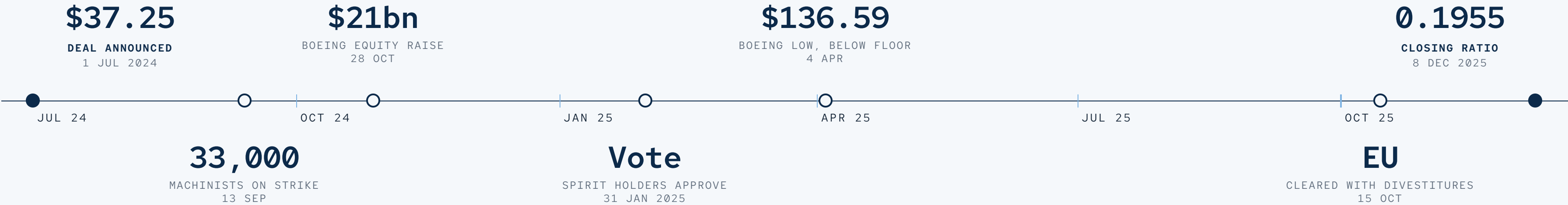
-\$0.48

cost of the collar
vs a fixed ratio

\$40.33

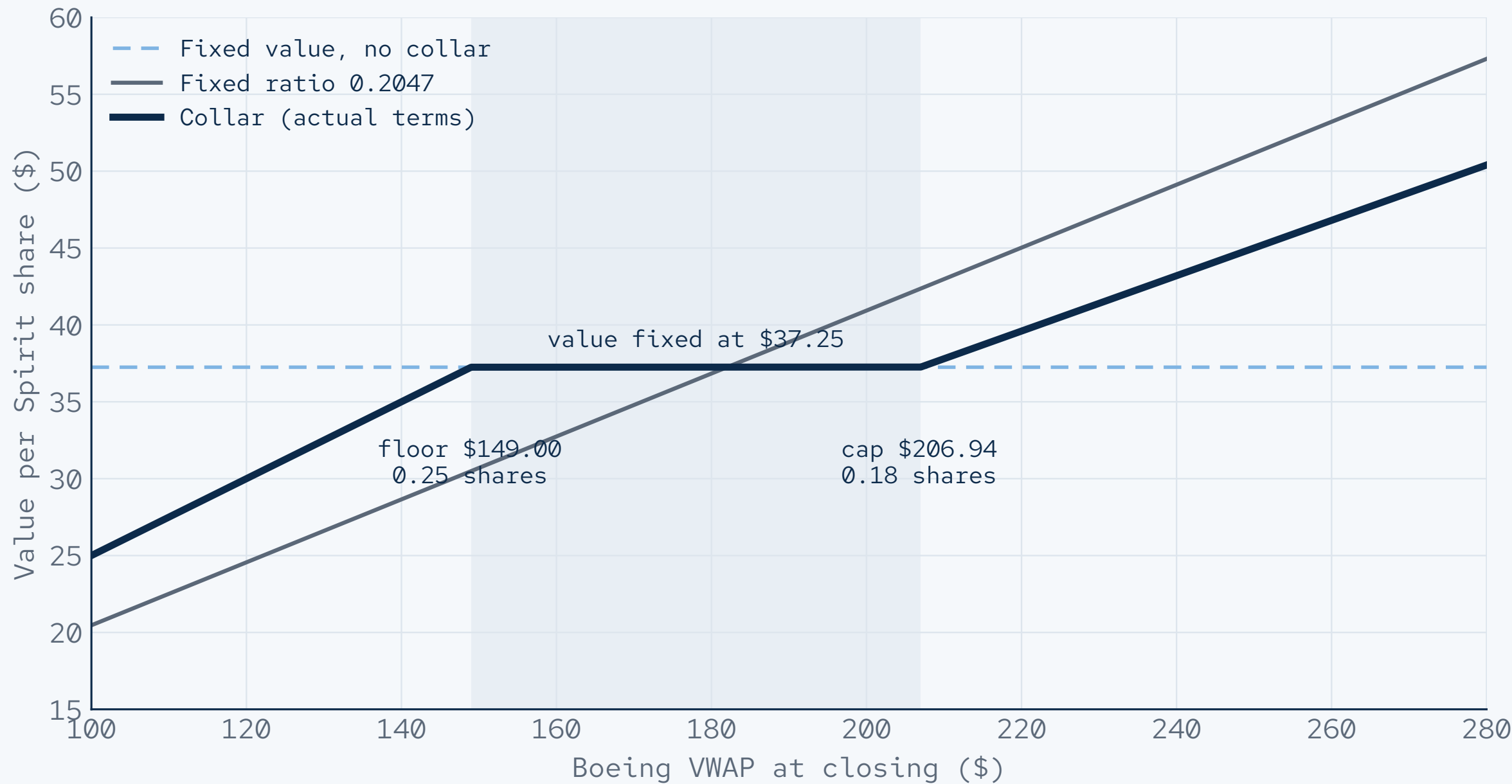
delivered at closing,
8 Dec 2025

FROM SIGNING TO CLOSING · SOLID = THE DEAL · HOLLOW = WHAT HAPPENED TO BOEING



Protection only between \$149 and \$207, then Boeing risk again.

VALUE RECEIVED PER SPIRIT SHARE, BY BOEING'S 15-DAY VWAP AT CLOSING



FLOOR · RATIO 0.25

\$149.00

18% below Boeing's \$182.01 close before signing

CAP · RATIO 0.18

\$206.94

only 14% above it

INSIDE THE BAND

\$37.25

shares adjust: \$37.25 / VWAP

IN OPTION TERMS

$\$37.25 - 0.25 P_{149} + 0.18 C_{206.94}$

Spirit holders sell puts and keep calls on Boeing

Real Boeing quotes in, the value of every term out.

#	STEP	WHAT IT DOES	OUTPUT
1	Deal terms	Floor, cap, ratios and VWAP window from the proxy; final ratio from the 8-K	0.1955 at closing
2	Quotes	Boeing end-of-day chains, bid and ask, 3 to 4 expiries a day	41,166 on 348 days
3	Implied vols	Black-76 on out-of-the-money options; no dividend, put-call-parity spot	13,944 kept
4	SVI per expiry	Arbitrage-free raw SVI; flat ATM vol beyond 7 weeks	0.54 pts RMSE
5	Collar value	Bond – 0.25 puts + 0.18 calls at their strike vols; Monte Carlo for averaging	daily, to closing

Boeing has paid no dividend since 2020, so the forward is $\text{spot} \times e^{rT}$ and American calls are worth their European value.

PUT VOL AT \$149 · 1 YEAR

33.6%

28 June 2024 surface, skew included

CALL VOL AT \$206.94 · 1 YEAR

32.0%

at-the-money 32.2%

BOEING REALISED · PRIOR YEAR

31.0%

36.9% over the 17 months that followed

DAILY SURFACES FITTED

336

AVERAGE FIT ERROR

0.54 pts

VWAP REBUILT VS ACTUAL

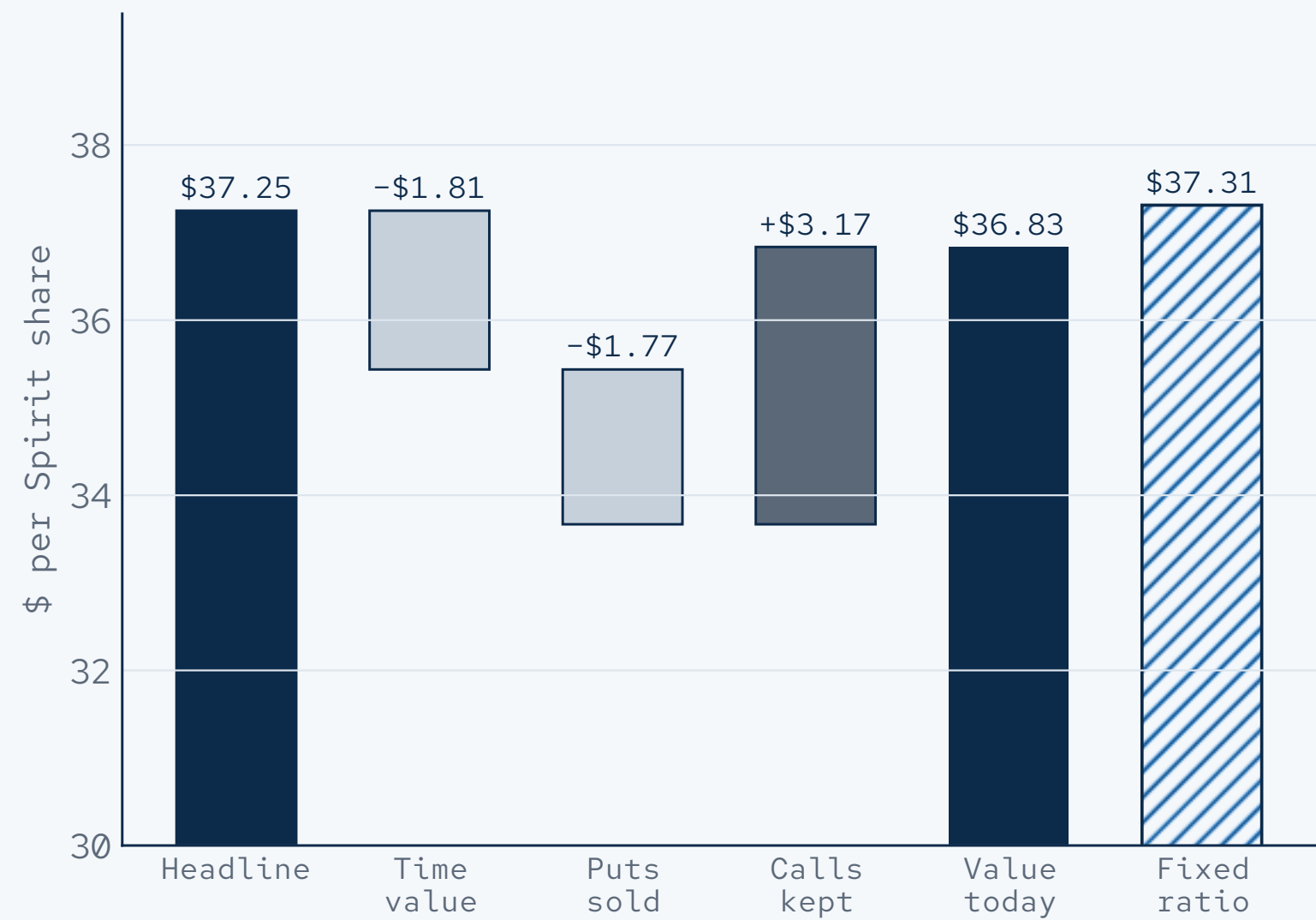
\$190.74 / \$190.54

EXCEL VS PYTHON

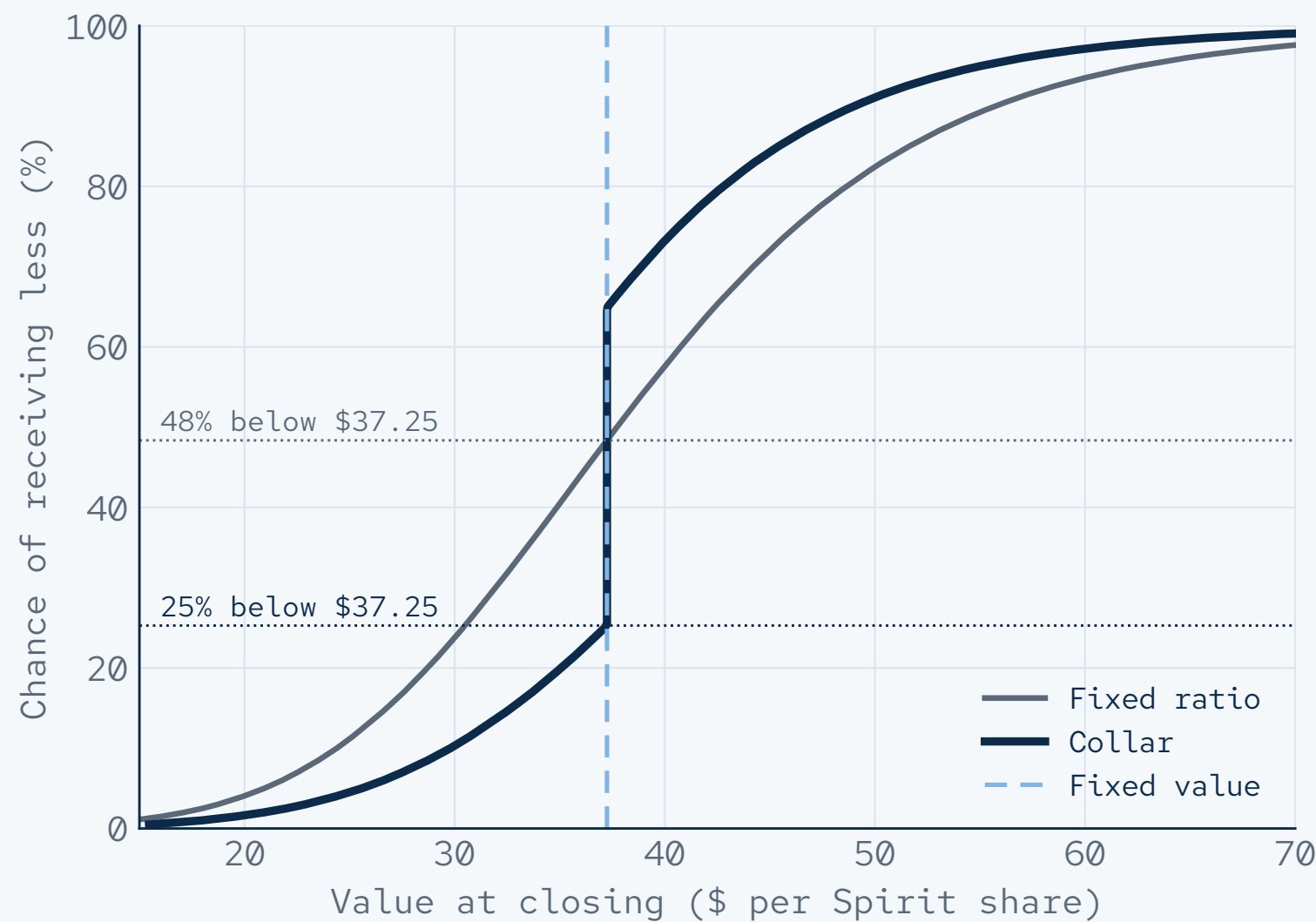
11 / 11 checks

Worth \$36.83: \$0.48 less than a fixed ratio, half the downside.

FROM HEADLINE TO VALUE · PER SPIRIT SHARE · ROUNDED TO THE CENT



CHANCE OF RECEIVING LESS THAN A GIVEN VALUE AT CLOSING



COLLAR VS FIXED RATIO

\$36.83 vs \$37.31

CHANCE BELOW \$37.25

25% vs 48%

WORST 5% OF OUTCOMES

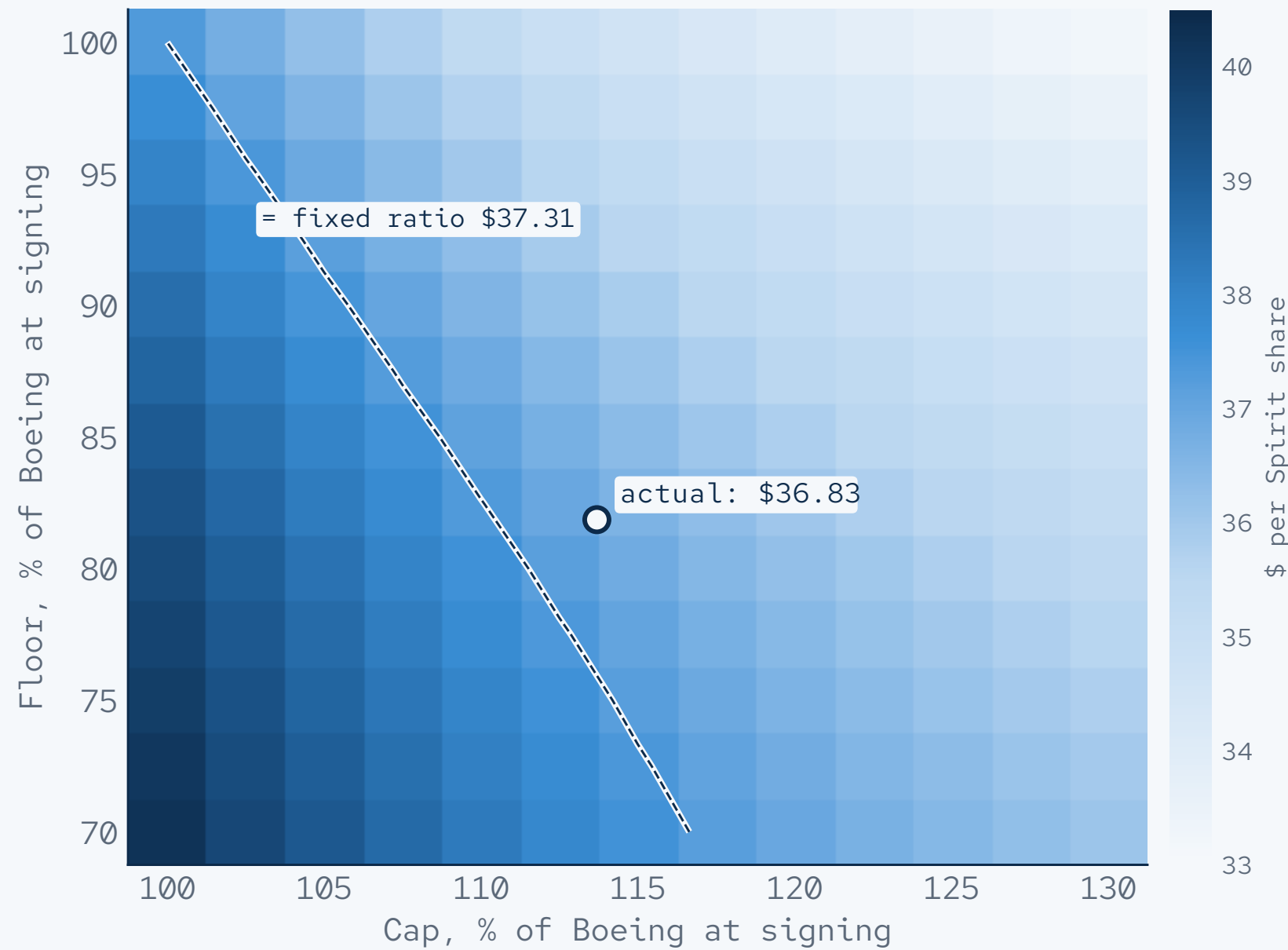
\$25.58 vs \$20.94

FIXED VALUE, NO COLLAR

\$35.44

A cap near \$201 and the collar would have cost nothing.

COLLAR VALUE BY FLOOR AND CAP · LINE = VALUE OF A FIXED RATIO



CAP THAT MATCHES A FIXED RATIO

\$201

instead of \$206.94, with the floor at \$149

FLOOR THAT MATCHES A FIXED RATIO

\$138

instead of \$149.00, with the cap at \$206.94

FLOOR AT \$140 · CAP AT \$200

+\$0.40 · +\$0.57

per Spirit share, each on its own

VALUE GIVEN UP VS A FIXED RATIO

≈ \$60M

\$0.48 × ~126M Spirit shares

Boeing broke the floor in 2024 and again in 2025.

BOEING DAILY CLOSE AGAINST THE COLLAR BAND · JUNE 2024 TO DECEMBER 2025



CLOSES BELOW THE FLOOR

16 of 362

autumn 2024 and spring 2025

BOEING LOW

\$136.59

4 April 2025; a closing then paid \$29.44

REALISED VS IMPLIED VOL

36.9% vs 32%

the insurance was cheap

MERGER-ARBITRAGE SPREAD

\$1.33

average; 85% implied completion if a break meant \$28.60

Accept the collar, but negotiate the band.

WHAT A SPIRIT SHARE RECEIVED · 8 DECEMBER 2025

OUTCOME	BOEING SHARES	VALUE	VS \$37.25	VS UNAFFECTED \$28.60
Collar (actual)	0.1955	\$40.33	+\$3.08	+41%
Fixed exchange ratio	0.2047	\$42.22	+\$4.97	+48%
Fixed value, no collar	0.1955	\$40.33	+\$3.08	+41%
Collar, if closed 4 Apr 2025	0.2155	\$29.44	-\$7.81	+3%

Boeing at \$206.27 on the closing day. The VWAP (about \$190.5) fell inside the band, so the collar and a plain fixed value gave the same ratio. The 4 April line shows the lag: ratio set on a 15-day average, shares delivered two days later.

CONCLUSION

The collar was fair insurance: \$0.48 a share for halving the chance of receiving less than \$37.25. Its weakness was the band, 18% of protection below and only 14% of room above. The board should have asked for a cap near \$201 and a walk-away below the floor, where Boeing's real risk sat.

Quotes on 348 days41,166

Value at signing\$36.83

Ratio rebuilt vs actual0.1953 / 0.1955

Deliverablesweb page · notebook · Excel · memo · deck